

ESC quality indicators for post-myocardial infarction care: transition and chronic coronary syndrome phases

Developed in collaboration with the European Association of preventive cardiology of the ESC

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Aims

We aimed to develop the European Society of Cardiology (ESC) quality indicators (QIs) for myocardial infarction (MI), from 1 year after hospital discharge, corresponding to transition to the chronic coronary syndrome phases.

Methods and results

We collaborated with the European Association of Preventive Cardiology (EAPC) and developed QIs for the long-term management of patients following MI. We applied the ESC methodology for QI development by (i) determining key domains of post-MI care; (ii) developing candidate QIs by performing a systematic review of the literature, and (iii) selecting the final set of QIs using a modified Delphi approach. In total, 18 QIs were identified across seven domains of care including (i) structural framework, (ii) risk assessment and follow-up, (iii) pharmacological management, (iv) rehabilitation, behavioural, and preventive interventions, (v) coronary revascularization, (vi) clinical outcomes, and (vii) patient-reported outcomes.

Conclusion

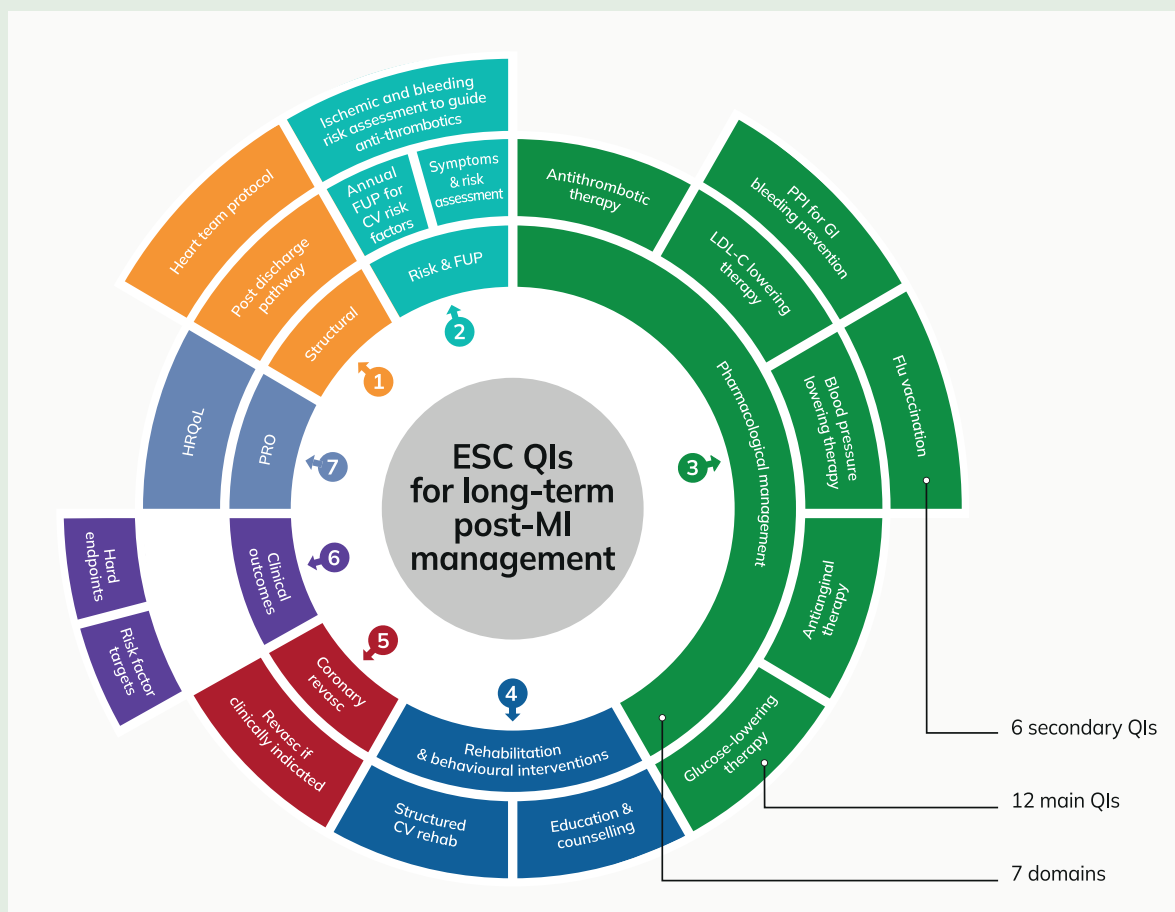
We present the ESC QIs from 1 year after hospitalization for MI, to standardize and address gaps in care for this high-risk group. These QIs are supported by evidence from contemporary literature, endorsed by expert consensus, and aligned with the 2024 ESC guidelines on the management of chronic coronary syndromes.

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Lay summary Measures to evaluate and improve the long-term management of patients following a heart attack are needed. In this paper, we identified key aspects of care that can help clinicians, decision-makers and patients improve the quality of care, from one year after a heart attack onwards, and help address inequalities and variations in clinical practice.

Graphical Abstract



Keywords Myocardial infarction • Secondary prevention • Quality of care • Guidelines • Implementation sciences • Quality improvement

Introduction

Myocardial infarction (MI) is associated with a high risk of subsequent major adverse cardiovascular events (MACE).¹ Prognosis is strongly influenced by the quality of clinical care and the adherence to guideline recommendations during the acute phase of MI, but also during long-term follow up.² Whilst studies have shown the benefits of secondary prevention programmes following MI,³ such as cardiac rehabilitation and multidisciplinary team management,⁴ there are still substantial gaps in quality of care and implementation of guideline recommendations, creating a need for interventions to address these gaps.^{5–7}

Healthcare systems around the world face financial challenges and shortages in resources, calling for a prioritization of quality improvement interventions that most strongly associated with favourable outcomes. Quality indicators (QIs) have been used to evaluate quality of care based on key management aspects that measure the achievement of valid care

processes, in line with available evidence and contemporary guidelines. The European Society of Cardiology (ESC) has a track record in the development of QIs for several cardiovascular conditions.^{8–11} Recently, the ESC QIs for patients with acute coronary syndromes were published following the development of the respective clinical practice guidelines.^{1,9} However, QIs for the long-term care of MI survivors during the transition from hospital-managed care during the first 12 months to community-managed care and chronic coronary syndrome phases have not been developed.

In collaboration with the European Association of Preventive Cardiology (EAPC), the group was tasked to develop a set of QIs that encompasses the spectrum of post-MI care. We used the ESC methodology for the development of QIs and ensured alignment with the most recent ESC guidelines.^{12,13} We anticipate that this set of QIs will be useful to evaluate adherence to guidelines and prioritize the domains where quality improvement initiatives could improve long-term outcomes after MI.¹⁴

Methods

The development of QIs was based on the principles of ESC methodology: (i) identification of key domains of care for the management of chronic coronary syndrome patient (specific phenotype: prior MI more than 1 year ago), (ii) selection of candidate QIs by constructing a systematic review of the literature, (iii) endorsement of the final set of QIs using a modified Delphi method, and (iv) feasibility evaluation of the selected QIs.¹² Following the results of the Delphi method, the QIs were classified as main and secondary. Main QIs are typically considered for the measurement of system performances, whereas main and secondary indicators can be used for quality improvement initiatives.¹²

Members of the working group

The group comprised Task Force members of previous ESC guidelines, in addition to representatives from the EAPC.^{9,12,13} Selection of the working group was based on diversity of expertise in the management of acute coronary syndromes (ACS) and chronic coronary syndrome, as well as in the development of QIs and ESC guidelines. A series of virtual meetings were convened among working group members between November 2023 and June 2025 before proceeding to Delphi voting.

Target population and domains of care

The target population for this set of QIs was defined as adults beyond the initial 12 months after an MI, taking into consideration the transition from the acute coronary syndrome phase to the stabilized phase, according to the 2024 ESC guidelines for the management of chronic coronary syndromes.¹³ The target population differs from patients with acute coronary syndromes, for whom a dedicated set of QIs have already been published.^{9,15} The key domains of care were constructed considering the setting of care, the typical interventions provided for this group of patients (pharmacological, rehabilitation, and behavioural interventions, coronary revascularization) and the relevant clinical outcomes for post-MI patients. Specifications were determined for each QI including a numerator, i.e. the set of patients who were delivered the QI, and a denominator, i.e. the set of patients who were eligible to receive the QI.

Systematic review

Search strategy

We performed a systematic review following the Preferred Reporting Items for Systematic Review and Meta-analyses statement. We searched publications in MEDLINE via Ovid and EMBASE using Embase.com. The search strategy was developed using keywords and medical subject heading (MeSH) terms and applying the Population, Intervention, Comparisons and Outcome framework to the post-MI context. In addition, the search was restricted to studies with a randomized controlled trial (RCT) design, conducted in humans and published in English. A similar approach was used in previous developments of QIs for patients with cardiovascular disease.^{8,10} The full search strategy is provided in [Supplementary material online, Table S1](#).

Eligibility criteria

We included studies that had the following criteria: (i) study population with at least 50% of post-MI adult patients (≥ 18 years old), (ii) evaluation of an intervention (follow-up strategy, pharmacological therapy, rehabilitation, education, prevention) on outcomes of interest, (iii) long-term outcome measures (relevant beyond 12 months), including hard endpoints (e.g. mortality, hospitalization), recommended targets for cardiovascular risk factors (e.g. lipids, diabetes, blood pressure, body mass index, smoking, alcohol, nutrition, physical activity), patient-reported outcomes (psychosocial well-being/health-related quality of life); (iv) clear definitions of interventions and outcome measurements; and (v) peer-reviewed RCT design. There was no restriction regarding sample size or follow-up duration.

Study selection

An internet-based review production software (Covidence, Swiss Academic Software, Switzerland) was used for study screening including duplicate removal and data management. Two reviewers (C.F. and A.A.) independently examined the abstracts of the studies against prespecified criteria for inclusion. Disagreements were resolved through a full-text review of the article or by discussions in a larger core group (B.G. and S.A.).

Data extraction

Publications that met the eligibility criteria were included in the initial phase of the review to ensure that the list of potential QI candidates was exhaustive and representative of the wide spectrum of post-MI patient care. The full texts of the included articles were then reviewed by a core group (B.G., S.A., C.F., and X.R.) to formulate candidate QIs.

Clinical practice guidelines and existing QIs

We reviewed the 2023 ESC guidelines for the management of acute coronary syndromes, the 2024 ESC guidelines for the management of chronic coronary syndromes and the 2025 ACC/AHA guidelines for the management of patients with acute coronary syndromes.^{1,13,16} Previously published and pertinent QIs^{8,9,15} were evaluated, as well as recommendations from the above guidelines against ESC criteria for QIs.

Data synthesis

Modified Delphi process

During the initial phase of the process, we formed candidate QIs which were then proposed to the wider group using the modified Delphi method.¹² The ESC criteria for QI development were presented to standardize the selection process. Each proposed QI was individually graded by each panel member through an online questionnaire using a nine-point ordinal scale for both validity and feasibility. We conducted two rounds, each concluding with a videoconference to discuss the results of the vote, confirm the selected choices and implement the suggested revisions for QI formulation.

Analysing voting results

The nine-point ordinal scale used in voting in each QI was interpreted as follows: (i) a score from 1 to 3 was considered non valid/feasible, (ii) a score from 4 to 6 was considered of uncertain validity/feasibility, and (iii) a score from 7 to 9 was considered as valid and feasible.¹² For each candidate QI, the median and mean deviation from the median were calculated to evaluate the distribution of votes. Indicators with median scores ≥ 7 for validity and ≥ 4 for feasibility in the second round were included in the final set of QIs. The main QIs were those with minimal dispersion (spread ≤ 1.5 for both validity and feasibility); secondary QIs were those with some dispersion (spread ≤ 1.5 for validity or feasibility).

Results

Domains of post-MI care

Seven domains of care for post-MI patients were identified. These included (i) structural framework, (ii) risk assessment and follow-up, (iii) pharmacological management, (iv) rehabilitation, and behavioural interventions, (v) coronary intervention/revascularization, (vi) clinical outcomes, and (vii) patient-reported outcomes ([Figure 1](#)). A checklist summarizing the domains of care and the selected QIs is provided in the [Supplementary material online, Figure S1](#).

Systematic review results

The literature review retrieved 5506 articles, of which 152 met the inclusion criteria ([Figure 2](#)). In total, 103 candidate QIs were extracted and subsequently included in the first Delphi round.

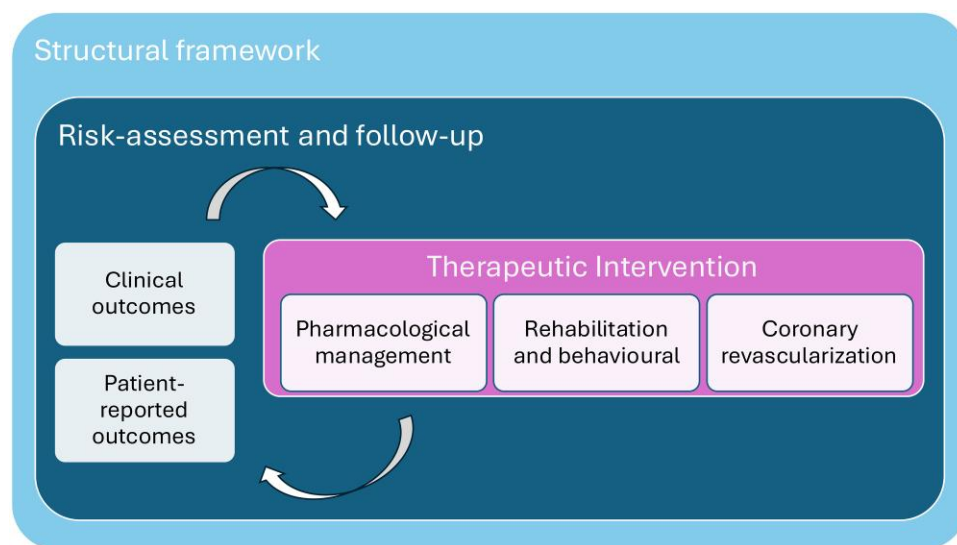


Figure 1 Interaction among the seven key domains of ESC quality indicators for myocardial infarction care, from one year after hospital discharge. The outer box indicates the structural domain, with the inner boxes comprising all other domains.

Modified Delphi results

Following the first round of voting, 21 candidate QIs were excluded due to poor feasibility and/or validity and 82 were revised for the second round based on the comments collected during the first stage of voting. Although QIs dedicated to HF management were considered feasible and valid, they were out of scope of this endeavour. We referred to the published QI documents for the management of heart failure and acute coronary syndromes for those overlapping situations.^{9,11}

A total of 18 main QIs were selected for the second round of voting, and finally 12 were classified as main QIs, and six as secondary QIs. Of note, one QI was initially excluded during the individual voting, but since this excluded QI was the only one in the interventional management domain, further discussion took place among voters to reach an acceptable version of this QI for the final version.

Quality indicators (Table 1)

Domain 1: structural framework

This structural domain refers to the institutional framework within which evidence-based care can be implemented following an MI. The initiation of a structured pathway early after discharge is well-known to improve outcomes in post-MI patients.^{1,16} The main QI pertains to healthcare centres managing post-MI patients with an established post-discharge pathway for the long-term management of cardiovascular risk factors, such as referral to cardiac rehabilitation and follow-up by healthcare professionals with an interest in post-MI management (Q1). The secondary QI refers to the requirement of multidisciplinary heart team protocols, including access to a tertiary centre for coronary revascularization and subsequent treatment (Q2). The general requirements for centre infrastructure were previously defined in the position statement of the EAPC on secondary prevention and rehabilitation.⁴

Domain 2: risk assessment and follow-up

Two main QIs and one secondary QI were selected in this domain to standardize the follow-up process. The first main QI specifies the need

for an annual follow-up with a healthcare professional to focus on the management of secondary prevention. The list of risk factors was defined based on the 2024 guidelines for the management of chronic coronary syndromes and includes modifiable risk factors treatable with pharmacological and non-pharmacological interventions (Q3). The second main QI emphasizes the evaluation of angina or equivalent symptoms despite optimal medical therapy, with non-invasive tests to evaluate the progression of coronary disease and its functional impact (Q4). The secondary QI refers to the individualized assessment of antithrombotic strategies considering the balance between individual ischaemic and bleeding risks (Q5). The criteria for ischaemic risk were based on the 2024 guidelines for the management of chronic coronary syndromes and for bleeding risk using the PRECISE-DAPT or ARC-HBR scores.^{13,17,18} In addition, the SMART score is typically recommended to evaluate residual risk in post-MI patients.¹⁹

Domain 3: pharmacological management

The pharmacological management domain comprises five main QIs and two secondary QIs. The first main QI is related to the long-term use of antiplatelet therapy either with aspirin or with clopidogrel (Q6). The second main QI refers to optimization of lipid-lowering by up-titration of statin intensity and/or statin combination with additional therapies if LDL-C targets are not reached (LDL-C below 1.4 mmol/L (55 mg/dL) and $\geq 50\%$ reduction from baseline level) after hospital discharge (Q7). The third main QI refers to the prescription of blood pressure lowering treatment in those with established hypertension. Essential hypertension is defined according to the 2024 ESC Guidelines for the management of elevated blood pressure and hypertension (Q8).²⁰ The fourth main QI includes the prescription of therapies to reduce anginal symptoms if clinically indicated (Q9). The fifth main QI refers to the prescription of novel glucose-lowering drugs in patients with type 2 diabetes mellitus (T2DM) to reduce CV risk (Q10). The secondary QIs in this domain refer to the prevention of gastro-intestinal bleeding with the prescription of a proton pump inhibitor in case of increased bleeding risk conditions, as defined in the 2024 guidelines for the management of chronic coronary syndromes

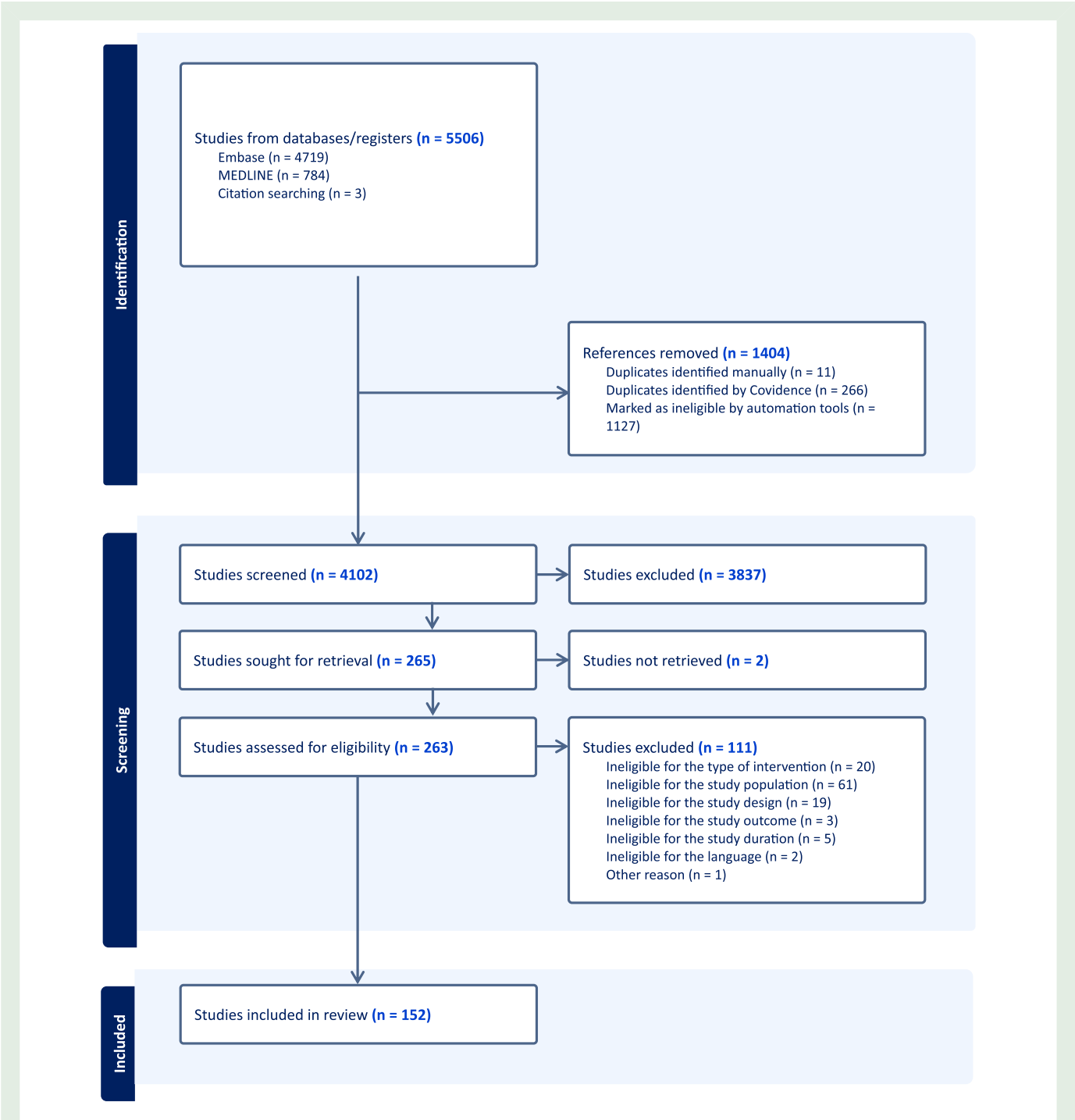


Figure 2 Preferred Reporting for Systematic Reviews and Meta-Analyses (PRISMA) flowchart for systematic reviews.

(Q11), and to the vaccination against influenza (Q12).¹³ Improving the adherence to secondary prevention medications is a continuous endeavour that can only be accomplished by collaborating with primary care providers and allied healthcare professionals (AHPs).

Domain 4: rehabilitation and behavioural interventions

This domain relates to rehabilitation and behavioural interventions with two main QIs. The first main QI underlines the importance of providing health-related education and counselling to improve adherence to

medications, healthy lifestyle, including dietary choices and psychological well-being (Q13). The second main QI is the participation in a multi-disciplinary cardiac rehabilitation programme (Q14). The secondary prevention and rehabilitation section of the EAPC have provided practical recommendations on the core components and goals of cardiac rehabilitation for specific cardiovascular conditions, including previous acute coronary syndrome.⁴ Cardiac rehabilitation programmes should be supervised by dedicated trained and responsible healthcare providers, such as cardiologists or AHPs experienced in implementing and delivering cardiac rehabilitation care.

Table 1 ESC quality indicators of MI care, from 1 year after hospital discharge**Domain 1. Structural framework**

(QI 1) Main 1.1 Percentage of healthcare centres* managing patients following MI that offer a comprehensive post-discharge pathway to manage long-term cardiovascular risk factors, with:

1. referral to a cardiac rehabilitation centre, and
2. long-term follow-up by a healthcare professional with experience in cardiovascular management.

Numerator: Number of health care centres managing patients following MI and offering a post discharge long-term pathway.

Denominator: Number of health care centres managing patients following MI.

(QI 2) Secondary 1.1 Percentage of healthcare centres* managing patients with previous MI that have multidisciplinary heart team protocols (or access to a specialized tertiary centre) to discuss appropriate subsequent treatment and coronary revascularization if required.

Numerator: Number of health care centres managing patients following MI with heart team protocols.

Denominator: Number of health care centres managing patients following MI.

*The requirements for centre infrastructure are defined in the position statement of the secondary prevention and rehabilitation section of the EAPC (European Journal of Preventive Cardiology (2021) 28, 496–509).⁴

Domain 2. Risk assessment and follow-up

(QI 3) Main 2.1 Percentage of patients following MI who have an annual follow-up with a healthcare professional for the management of their risk factors*.

*The risk-factors defined in the 2024 guidelines for the management of chronic coronary syndromes are:¹³

1. Smoking and substance abuse,
2. Obesity and being overweight,
3. Hyperlipidaemia,
4. Diabetes,
5. Arterial hypertension,
6. Non-Mediterranean diet and excessive alcohol consumption and
7. Low levels of physical activity and exercise

Numerator: Number of post-MI patients who have an annual follow-up with a healthcare professional for the management of risk factors.

Denominator: Number of post-MI patients.

(QI 4) Main 2.2 Percentage of patients following MI with angina or equivalent symptoms despite optimal medical therapy who undergo ECG, echocardiography and either non-invasive imaging or angiography.

Numerator: Number of post-MI patients with angina or equivalent symptoms despite optimal medical therapy who undergo ECG, echocardiography, and either non-invasive imaging or angiography.

Denominator: Number of patients following MI with angina or equivalent symptoms despite optimal medical therapy.

(QI 5) Secondary 2.1 Percentage of patients following MI who have an individualized assessment of their risk of recurrent ischaemic and bleeding events* to guide antithrombotic strategy.

*Ischaemic risk factors include prior stenting of the left main stem, proximal LAD or last remaining patent artery; suboptimal stent deployment; stent length of >60 mm; diabetes mellitus; CKD; bifurcation with two stents implanted; treatment of chronic total occlusion; and previous stent thrombosis on adequate antithrombotic therapy. Bleeding-risk criteria according to PRECISE-DAPT or ARC-HBR scores.¹³ The SMART-ESC score is also typically recommended to evaluate the residual risk in post-MI patients.

Numerator: Number of patients following MI who have an individualized assessment of their risk of recurrent ischaemic and bleeding events* to guide antithrombotic strategy.

Denominator: Number of patients following MI.

Domain 3. Pharmacological management

(QI 6) Main 3.1 Percentage of patients following MI with no indication for anti-coagulation who are prescribed life-long antiplatelet monotherapy with either aspirin or clopidogrel in the absence of contraindication.

Numerator: Number of patients following MI with no indication for anti-coagulation who are prescribed life-long antiplatelet monotherapy with either aspirin or clopidogrel in the absence of contraindication.

Denominator: Number of post-MI patients with no indication for anti-coagulation.

(QI 7) Main 3.2 Percentage of patients following MI with LDL-C levels above the targets (>1.4 mmol/L [> 55mg/dL] and <50% reduction) despite high or maximally tolerated intensity statin who receive an additional LDL-C lowering therapy.

Numerator: Number of patients following MI with LDL-C levels above the targets (>1.4 mmol/L and <50% reduction) despite high or maximally tolerated intensity statin who receive an additional LDL-C lowering therapy.

Denominator: Percentage of patients following MI with LDL-C levels above the targets (>1.4 mmol/L and <50% reduction) despite high or maximally tolerated intensity statin.

(QI 8) Main 3.3 Percentage of patients following MI with essential hypertension who are prescribed pharmacological and/or non-pharmacological blood pressure lowering treatment.

Numerator: Number of patients following MI with essential hypertension who are prescribed pharmacological and/or non-pharmacological blood pressure lowering treatment.

Denominator: Number of patients following MI with essential hypertension.

(QI 9) Main 3.4 Percentage of patients following MI presenting angina or equivalent symptoms* who are prescribed therapies to relieve angina symptoms (e.g. short acting nitrates, beta-blockers or calcium antagonists).

*Symptom characteristics are described in the 2024 guidelines for the management of chronic coronary syndromes.¹³

Numerator: Number of patients following MI with angina or equivalent symptoms who are prescribed therapies to relieve anginal symptoms (e.g. short acting nitrates, beta-blockers or calcium antagonists).

Denominator: Number of patients following MI with angina or equivalent symptoms.

(QI 10) Main 3.5 Percentage of patients following MI with T2DM who are treated with GLP1-RA and/or SGLT2 inhibitors with proven cardiovascular benefits in the absence of contraindication.

Numerator: Number of patients following MI with T2DM who are treated with GLP1-RA and/or SGLT2 inhibitors with proven cardiovascular benefits in the absence of contraindication.

Denominator: Number of patients following MI and T2DM.

(QI 11) Secondary 3.1 Percentage of patients following MI at increased risk of gastrointestinal bleeding* who are prescribed a proton pump inhibitor with antithrombotic therapy.

*Increased risk of gastrointestinal bleeding defined in the 2024 guidelines for the management of chronic coronary syndromes.¹³

Numerator: Number of patients following MI at increased risk of gastrointestinal bleeding who are prescribed a proton pump inhibitor with antithrombotic therapy.

Denominator: Number of patients following MI on antithrombotic therapy who are at increased risk of gastrointestinal bleeding.

(QI 12) Secondary 3.2 Percentage of patients following MI who are vaccinated against influenza.

Numerator: Number of patients following MI vaccinated against influenza.

Denominator: Number of patients following MI.

Domain 4. Rehabilitation and behavioural interventions

(QI 13) Main 4.1 Percentage of patients following MI who receive annual education and counselling to improve adherence to medications, healthy lifestyles* and psychological well-being.

*Healthy lifestyles, including smoking and substance abuse cessation, weight control, Mediterranean diet, limited alcohol consumption, physical activity, and exercise, are defined in the 2024 guidelines for the management of chronic coronary syndromes.¹³

Numerator: Number of patients following MI who receive annual education and counselling to improve adherence to medication, healthy lifestyles and psychological well-being.

Denominator: Number of patients following MI.

(QI 14) Main 4.2 Percentage of patients following MI who participate in a structured multidisciplinary cardiac rehabilitation programme as defined in current recommendations*.

*Definition of a comprehensive cardiac rehabilitation programme refers to a 2020 position paper from the secondary prevention and rehabilitation section of the EAPC.⁴

Numerator: Number of patients following MI who participate in a structured multidisciplinary cardiac rehabilitation programme.

Denominator: Number of patients following MI.

Domain 5. Coronary intervention/revascularization

(QI 15) Main 5.1 Percentage of patients following MI with hemodynamically significant obstructive coronary artery disease and on-going angina despite maximum tolerated anti-anginal therapies, or with another indication for revascularization* who are assessed for possible coronary revascularization, preferably by a Heart Team.

*Indications for coronary revascularization based on the 2024 guidelines for the management of chronic coronary syndromes are:¹³

8. Left main disease

9. Multivessel disease

10. Single- or double- vessel disease involving the proximal LAD

11. Symptomatic single- or double- vessel disease not involving the proximal LAD

Numerator: Number of patients following MI with hemodynamically significant obstructive coronary artery disease and on-going angina despite maximum tolerated anti-anginal therapies, or with another indication for revascularization who are assessed for possible coronary revascularization, preferably by a Heart Team.

Denominator: Number of patients following MI with hemodynamically significant obstructive coronary artery disease and on-going angina despite maximum tolerated anti-anginal therapies, or with another indication for revascularization.

Domain 6. Clinical outcomes

(QI 16) Secondary 6.1 Percentage of patients following MI who are screened for annual rates of:

12. Death from any cause

13. Cardiovascular death

14. Cardiovascular hospitalization (myocardial infarction, stroke, unplanned coronary revascularization, major arrhythmia, critical limb ischaemia, or heart failure)

Numerator: Number of patients following MI who are screened for annual rates of clinical outcomes.

Denominator: Number of post-MI patients.

(QI 17) Secondary (composite) 6.2 Percentage of patients following MI who achieve *all relevant* goals of secondary prevention therapies as recommended by the 2024 guidelines for the management of chronic coronary syndromes¹³:

15. LDL-C level <1.4 mmol/L (<55 mg/dL) and a $\geq 50\%$ reduction from baseline level (applies to all)
16. HbA1c <7% (applies to patients with diabetes)
17. Systolic blood pressure below 129 mmHg (applies to patients with hypertension)
18. Use of pharmacological and behavioural strategies for smoking cessation (applies to smokers)
19. Self-reported aerobic physical activity or with the use of mobile health, defined as 150–300 min moderate activity per week and reduced sedentary time and engagement in light activity throughout the day (applies to all)
20. Self-reported diet high in vegetables, fruit and whole grains (Mediterranean-diet) (applies to all)
21. Self-reported limited alcohol consumption (<100 g per week) (applies to all)
22. Obtain or maintain a healthy weight (BMI 18.5–25 kg/m²)* (applies to all)

*Consider adjustment of normal BMI for Asian populations

Numerator: Number of patients following MI who achieve *all relevant* therapy goals.

Denominator: Number of patients following MI who are *eligible* for all the above therapy goals.

Domain 7. Patient-reported outcomes

(QI 18) Main 7.1 Percentage of patients following MI who are assessed annually for health-related quality of life as reported by the patients and/or structured questionnaires (e.g EQ-5D, HeartQoL, GAD, PHQ).

Numerator: Number of post-MI patients who are assessed annually for health-related quality of life as reported by the patients and/or structured questionnaires.

Denominator: Number of patients following MI.

Abbreviations: BMI, body mass index; CKD, chronic kidney disease; EAPC, European Association of Preventive Cardiology; EQ-5D, EuroQoL-5 Dimension; GAD, general anxiety disorder; GLP1-RA, glucagon-like peptide-1 receptor agonists; HeartQoL, health-related quality of life for patients with ischaemic heart disease; HbA1c, haemoglobin A1C; LAD, left anterior descending; LDL-C, low-density lipoprotein cholesterol; MI, myocardial infarction; PHQ, patient health questionnaire; SGLT2, sodium-glucose cotransporter 2; SMART, secondary manifestations of arterial disease risk score; T2DM, type 2 diabetes mellitus.

Domain 5: coronary intervention/revascularization

The only QI in the interventional management section refers to the clinical indications for coronary revascularization in post-MI patients. The key criteria are residual obstructive disease associated with anginal symptoms despite optimal medical therapy (**Q15**), or the presence of anatomic indications for coronary revascularization based on the 2024 guidelines for the management of chronic coronary syndromes.¹³ Coronary revascularization has shown consistent benefit in patients with anginal symptoms despite optimal medical therapy.²¹

Domain 6: clinical outcomes

Two secondary QIs were selected. The first defines the type of clinical outcome that needs to be screened annually to evaluate the prognosis in post-MI patients, such as mortality or cardiovascular hospitalizations (**Q16**). To capture this QI, clinical outcomes should be routinely collected using linked data from national registries. The second QI in this domain is a composite *opportunity-based* QI to promote the overarching aim to achieve pertinent targets from secondary prevention interventions for post-MI patients by combining the goals of therapies in a single QI (**Q17**). The 2024 guidelines for the management of chronic coronary syndromes provided recommendations for lifestyle counselling and intervention in patients with established chronic coronary syndromes.¹³

Domain 7: patient-reported outcomes

The QI in this domain refers to the annual assessment of patient reported health-related quality of life using structured and validated questionnaires, such as the EuroQoL-5-Dimension (EQ-5D) and the health-related quality of life for patients with ischaemic heart disease (HeartQoL) (**Q18**). EQ-5D and HeartQoL are typical questionnaires used in previous studies. However, other instruments can also be considered, especially for mental health, such as the generalized anxiety disorder or the patient health questionnaire (PHQ).²²

Discussion

This document presents the ESC QIs for MI from one year after hospital discharge, corresponding to the transition and chronic coronary syndrome phases. The selection of QIs was based on a systematic review of original articles and aligned with the 2024 ESC guidelines on the management of chronic coronary syndromes.¹³ This document was developed in collaboration with the EAPC. The participation of guideline Task Force members, along with experts from diverse domains, aimed to combine the validity of selected QIs with feasibility in routine clinical practice.^{4,13} The QIs can be used by both healthcare professionals and regulators to systematically assess quality of care over time and across geographical areas.

In contrast to the recently published ESC QIs for the management of ACS,⁹ this paper pertains to the quality of long-term management from 1 year after hospital discharge, following MI. The current suite of QIs integrates with those intended for ACS, heart failure, and other ESC QIs.^{9,11} Whilst there is no plan to embed these QIs in any specific ESC guideline, their development was aligned with existing clinical practice guidelines and complements published position papers and QIs on secondary prevention and cardiac rehabilitation.^{4,13} The current document is dedicated mainly to healthcare providers and centres involved in the long-term management of post-MI patients in the outpatient setting.⁴ This paper considers the pathway of post-MI patients as a continuum, transitioning from the inpatient to the fully out-patient, community-managed setting, and constitutes a standardized approach for the evaluation of guideline adherence and quality improvement initiatives in secondary prevention.⁴ The following domains were considered to be relevant to monitor the quality of care of post-MI patients: (i) structural framework, (ii) risk assessment/follow-up, (iii) pharmacological management, (iv) rehabilitation/behavioural interventions, (v) coronary intervention/revascularization, (vi) outcomes, and (vii) patient-reported outcomes.

The developed QIs can be used to assess the quality of care of post-MI patients using routinely collected data. Such efforts would identify not

only gaps in care delivery but also missing key data variables in existing registries and electronic health records. The current effort is focused primarily on post-MI, as this group comprises a substantial portion of patients with chronic coronary syndromes and faces a high risk of adverse outcomes.²³ Moreover, patients with a previous MI are in general well-represented in studies of preventative interventions. Post-MI is an area of care where new preventive therapies or targets are considered by the guidelines, such as novel lipid-lowering therapies, novel glucose-lowering drugs, or updated blood pressure targets.^{20,24,25} The feasibility of capturing QIs pertinent to pharmacological treatment was deemed to be high, given that such information tends to be routinely collected in cardiovascular databases. In addition, evidence from registries suggests that intensification of recommended preventive pharmacological therapies can result in improvement of clinical outcomes in post-MI patients.²⁶

There is also a strong emphasis on non-pharmacological interventions, including behavioural, educational, and rehabilitation components. The QIs for lifestyle modification in the risk mitigation process of post-MI patients remain as important as those described in the pharmacological management domain. Such aspects were dealt with lifestyle recommendations on smoking, alcohol consumption, physical activity, nutrition, sleep, weight management, and psychosocial aspects.²⁷ The importance of centre-based cardiovascular rehabilitation is also mentioned in previous recommendations.⁴ In addition to the management of cardiovascular risk factors this paper includes QIs on long-term antithrombotic strategies and coronary revascularization, according to the clinical indications described in the chronic coronary syndrome guidelines.¹³

Real-world studies have shown delays in the implementation of new evidence, resulting in missed opportunities for patients who do not achieve the up-to-date recommended targets for cardiovascular risk factor control (e.g. LDL-C, systolic blood pressure, body mass index, and HbA1c) or are not treated according to the standards of care.^{5,28} The issues of inertia and the lack of coordination have been well described in the literature.^{28,29} This document aims to standardize the care process of post-MI patients into valid, feasible, and implementable QIs, thereby improving their quality of care and subsequent outcomes. Although gaps in care are multifactorial, they are partially due to structural factors where access to, and processes of, healthcare systems are suboptimal. For instance, the shortage of cardiac rehabilitation facilities is one of the major causes for suboptimal care following MI.³⁰ The proposed QIs for post-MI patients can catalyse the prioritization of preventive and therapeutic efforts to improve clinical outcomes.³¹

Study limitations

Although the identification and formulation of the proposed QIs were based on a rigorous methodology, the voting and final selection were decided by a panel of experts using the Delphi approach. There might be subjectivity in the final selection, but this approach is intrinsic for most documents elaborated by medical societies.³² The threshold for the definition of main vs. secondary QIs was predefined prior to the results of the voting, and there was large consensus favouring the voting outcomes during subsequent deliberation of the results. Of note, one of the selected QIs (the one related to coronary intervention/revascularization domain) was initially during the first round but subsequently revised with a consensus to integrate it in the final document given the emphasis of such aspect of care in the 2024 guidelines on the management of chronic coronary syndromes.¹³ The QI on glucose-lowering therapy was initially classified as a secondary QI but was then reclassified as a main QI after a dedicated voting round during the modified Delphi process.

In addition to the validity of the QIs, the panel has also evaluated the feasibility of implementing those QIs in everyday practice. However,

further initiatives should integrate the suggested QIs into practice by defining strategies available in the local context. The post-MI QIs aim to audit practice and improve outcomes in daily managed patients by acknowledging the gap between optimal guideline-based treatment and actual care of post-MI patients.¹³ Future initiatives to improve QI attainment can be implemented based on local, regional, and global assessments of QIs.¹³

Conclusion

We present the ESC QIs for MI from one year after hospital discharge to standardize and address gaps in care for this high-risk group. These QIs are supported by evidence from contemporary literature, endorsed by expert consensus, and are aligned with the 2024 ESC guidelines on the management of chronic coronary syndromes.

Supplementary material

Supplementary material is available at *European Journal of Preventive Cardiology*.

Author contributions

Baris Gencer (Conceptualization [lead]; Data curation [lead]; Formal analysis [lead]; Investigation [lead]; Methodology [lead]; Project administration [lead]; Supervision [lead]; Validation [lead]; Visualization [lead]; Writing—original draft [lead]; Writing—review & editing [lead]), Roberto F. E. Pedretti (Conceptualization [equal]; Investigation [supporting]; Methodology [supporting]; Project administration [supporting]; Supervision [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Chris P Gale (Conceptualization [supporting]; Formal analysis [supporting]; Investigation [supporting]; Methodology [supporting]; Supervision [supporting]; Validation [supporting]; Writing—original draft [supporting]; Writing—review & editing [supporting]), Felicità Andreotti (Formal analysis [supporting]; Investigation [supporting]; Methodology [supporting]; Validation [supporting]; Visualization [supporting]; Writing—review & editing [supporting]), Christiaan Vrints (Conceptualization [supporting]; Formal analysis [supporting]; Investigation [equal]; Methodology [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Elena Cavarretta (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Annett Salzwedel (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Konstantin A. Krychtiuk (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Ingo Ahrens (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Constantinos H. Davos (Conceptualization [equal]; Investigation [supporting]; Methodology [supporting]; Project administration [supporting]; Supervision [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Giuseppe Biondi-Zoccai (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Demosthenes Panagiotakos (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Trine Moholdt (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Kostantinos Koskinas (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Matthias Wilhelm (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Alessandro Sionis (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Xavier Rossello (Conceptualization [equal]; Data curation [supporting]; Formal analysis [supporting]; Investigation [supporting]; Methodology [equal]; Validation [supporting]; Writing—review & editing [supporting]), Amr Abdelrahman (Data curation [supporting]; Writing—review & editing [supporting]), Cedric Follonier (Conceptualization [equal]; Data curation [equal]; Formal

analysis [supporting]; Methodology [equal]; Software [equal]; Writing—review & editing [supporting]), Paul Dendale (Investigation [supporting]; Validation [supporting]; Writing—review & editing [supporting]), and Suleman Aktaa (Conceptualization [lead]; Data curation [lead]; Formal analysis [lead]; Investigation [lead]; Methodology [lead]; Project administration [lead]; Supervision [lead]; Validation [lead]; Writing—review & editing [lead]))

Conflict of interest: G.B.-Z. has consulted, lectured and/or served as advisory board member for Abiomed, Advanced Nanotherapies, Amarin, AstraZeneca, Bamed, Cardionovum, Cepton, Crannmedical, Endocore Lab, Eukon, Guidotti, Innovheart, Menarini, Microport, Opsens Medical, Servier, Synthesa, Terumo, and Translumina, outside the present work. F.A. reports receiving speaker or consultancy fees from Amgen, Menarini, Novo Nordisk and Servier, unrelated to the present article. All other authors report no conflict of interest.

Data availability

Voting data can be shared upon request.

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